FOR A FLAT MAP OF NOETHERIAN RINGS THE COMPLETED EXT AGREE WITH THE EXT OF THE REDUCED MODULES

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness of M and B over A giving exactness of the reduced free resolution. used at proof lines 29393-29396. Is what makes the second complex compute Ext of the reduced modules.
- The two complexes computing Ext from a free resolution. used at proof lines 29397-29406. Identifies the objects on each side of the asserted equality.
- Artin-Rees for the modules of a complex, comparing the completed system with the system of reduced modules. invoked at proof lines 29410-29411. Converts the termwise identification into the equality of inverse limits.

Source reference. Original lemma 28867: sources/stacks-more-algebra/pinned-original-source.tex, source lines 28867–28867. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \to B$ be a flat homomorphism of Noetherian rings. Let $I \subset A$ be an ideal. Let M, N be A-modules. Set $B_n = B/I^nB$, $M_n = M/I^nM$, $N_n = N/I^nN$. If M is flat over A, then we have*

$$\lim \operatorname{Ext}^i_B(M, N)/I^n \operatorname{Ext}^i_B(M, N) = \lim \operatorname{Ext}^i_{B_n}(M_n, N_n)$$

for all $i \in \mathbf{Z}$.

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Proof. Choose a resolution

$$\ldots \to P_2 \to P_1 \to P_0 \to M \to 0$$

by finite free B-modues P_i. Set $P_{i,n} = P_i/I^nP_i$. Since M and B are flat over A, the sequence

$$\ldots \to P_{2,n} \to P_{1,n} \to P_{0,n} \to M_n \to 0$$

is exact. We see that on the one hand the complex

$$\mathrm{Hom}_B(P_0, N) \to \mathrm{Hom}_B(P_1, N) \to \mathrm{Hom}_B(P_2, N) \to \ldots$$

computes the modules $\mathrm{Ext}^i_B(M, N)$ and on the other hand the complex

$$\mathrm{Hom}_{B_n}(P_{0,n}, N_n) \to \mathrm{Hom}_{B_n}(P_{1,n}, N_n) \to \mathrm{Hom}_{B_n}(P_{2,n}, N_n) \to \ldots$$

computes the modules $\mathrm{Ext}^i_{B_n}(M_n, N_n)$. Since

$$\mathrm{Hom}_{B_n}(P_{i,n}, N_n) = \mathrm{Hom}_B(P_i, N)/I^n \mathrm{Hom}_B(P_i, N)$$

we obtain the result from Lemma 28867 part (2). □